# BANDHAN HALDER

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## PROFESSIONAL SUMMARY

A passionate Physical Design Engineer with a focus in Static Timing Analysis (STA) and timing signoff, currently working as a Synopsys contractor on UCIe IP. I have refined my skills by signing off 3 UCIe blocks, resolving timing violations, and optimizing interface timing for SoC integration. Hands on experience across the full backend flow (floorplanning, CTS, routing, ECO and timing closure) on advanced and mature process nodes (TSMC 5nm, 4nm, 3nm, 45nm, 180nm), using Synopsys and Cadence tools, with Python, Perl and Tcl automation. Pursuing an M.Sc. in Electrical and Electronic Engineering (EEE) at BRAC University alongside full-time work. Eager to leverage skills and contribute effectively on field of PD projects.

## PROFESSIONAL EXPERIENCE

**Physical Design Engineer (Contractor)** | Synopsys | June 2025 – Present

- Perform STA and timing signoff on 3 UCIe blocks (Both Gen1 16Gbps and Gen2 40Gbps) using PrimeTime, with StarRC parasitics, Library Compiler and MSIP checks, across corners and modes.
- Analyzed and closed setup, hold, clock transition, data transition and minimum pulse width (MPW) violations in each of the 3 blocks inside UCIe, identifying root causes and driving fixes through to clean signoff.
- Optimized interface timing for SoC integration: doing internal and interface timing analysis and detailed run-to-run comparisons to keep UCIe interface paths closed and consistent.
- Worked as the timing point of contact (PoC) by delivering ECOs that resolved MPW violations; tracked and managed issues and deliverables in JIRA, identify early flow inconsistencies and keep the flow stable for QoR (quality of results) runs.
- Prepare accurate timing reports and signoff presentations for design reviews with the team.
- Completed physical design verification (PDV) training and assisted team members with PDV analysis and fixes.
- Resolved timing and DRC issues during post-recruitment training blocks to deliver clean, violation-free designs.
- Develop Python, Perl and Tcl automation scripts to streamline flow execution, reporting and debug.

**Physical Design Engineer** | ULKASEMI | February 2025 – Present

- Execute the full place-and-route (PnR) flow using Cadence Innovus and Synopsys ICC2.
- Assisted in timing signoff for 2 ADI projects and supported senior engineers in debugging critical timing issues.
- Delivered ECO targets for a metal-only ECO that resolved setup violations while maintaining hold closure and overall timing integrity.

## CORE COMPETENCIES

- **Timing & Signoff:** Static Timing Analysis (STA), timing signoff, timing closure, setup and hold fixing, clock transition and data transition (slew) fixing, minimum pulse width (MPW) analysis, ECO (Engineering Change Order), metal-only ECO, SoC interface timing optimization, QoR analysis
- **Physical Design:** Floorplanning, clock tree synthesis (CTS), routing, place and route (PnR), parasitic extraction, DRC/LVS, physical design verification (PDV), UCIe Gen1/Gen2
- **EDA Tools:** Synopsys PrimeTime, ICC2, Fusion Compiler, StarRC, Library Compiler; Cadence Innovus, Virtuoso, Genus, Calibre; Sentaurus TCAD; ModelSim
- **Scripting & Other:** Python, Perl, Tcl, Bash, C, C++; Linux; JIRA; MATLAB, PSpice

## EDUCATION

**M.Sc. in Electrical and Electronic Engineering (EEE)** | BRAC University | In progress

- Focus: semiconductor devices and VLSI. Coursework: Semiconductor Devices; Fundamentals of Nanoelectronics Technology

**B.Sc. in Electrical & Electronic Engineering** | Ahsanullah University of Science & Technology (AUST) | 2024

- CGPA: 3.686/4.00; 4 semester scholarships in 8 semesters
- Thesis: Analysis of Structural & Electrical Properties of Samarium Doped Bismuth Ferrite

**Higher Secondary Certificate** | Government Science College | 2019 | GPA 5.00/5.00
**Secondary School Certificate** | Dhanmondi Govt. Boys' High School | 2017 | GPA 5.00/5.00

## ACADEMIC PROJECTS

**10T SRAM Cell Design**

- Led a team of 4 in designing 10T, 6T, 5T and 4T SRAM cells in Cadence Virtuoso (90nm), building schematics and layouts from a reference paper.
- Calculated power and area for each cell and achieved a clean DRC and LVS result, delivering on time within the academic schedule.

**Advanced Home Automation & Security: RTL-to-GDSII Full Chip**

- Led a team of 4 through the end-to-end flow from RTL to GDSII on 45nm using ModelSim, Genus and Innovus.
- Allocated tasks and tracked deadlines, met functional and timing goals and delivered a successful project demo.

## ADDITIONAL INFORMATION

- Industrial attachment: Neural Semiconductor and Bangladesh Satellite Company Limited (BSCL).
- Relevant coursework: VLSI 1 & 2, Digital Electronics, Fabrication and Processing, Electronics 1 & 2, Signals & Systems, Programming Language.